Fiscal Year:	FY 2005	**Task Last Updated:**	FY 05/26/2005
PI Name:	Hughes-Fulford, Millie Ph.D.		
Project Title:	Mechanisms and Functional Consequences of Protein Kinase C Isoform Translocation Inhibition in Monocytes Exposed to Microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	(1) Cell Culture		
PI Email:	milliehf1@gmail.com	**Fax:**	FY 415-750-6667
PI Organization Type:	NON-PROFIT	**Phone:**	415-750-6940
Organization Name:	Northern California Institute for Research and Education		
PI Address 1:	Lab of Cell Growth, Mail Code 151F, VAMC		
PI Address 2:	4150 Clement Street		
PI Web Page:			
City:	San Francisco	**State:**	CA
Zip Code:	94121	**Congressional District:**	8
Comments:	PI is deceased as of February 2021		
Project Type:	Flight	**Solicitation / Funding Source:**	2001 Space Biology ILSRA
Start Date:	08/15/2002	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	2
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	Pre/Post Flight		
Flight Assignment:	Soyuz Taxi/ISS KUBIK Flight; Renewed. Updated Aug. 26, 2005.		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC2-1361		
Performance Goal No.:			
Performance Goal Text:			
	This study is aims to investigate the mechanisms by which microgravity affects protein kinase C (PKC) signaling in monocytes and the functional consequences of these changes on downstream signaling events, gene expression and cell fate. When protein kinase C isoforms are activated they translocate to the cell membrane, which is an key step in signal transduction by these kinases. Previous spaceflight experiments have shown that protein kinase C isoform translocation in monocytes & lymphocytes is inhibited in microgravity, with individual isoforms being differentially affected. The objectives of this research project are: (1) Fully characterize the effect of microgravity on, (a) the activation of key PKC isoforms (b) spatial distribution / redistribution of PKC isoforms within the cell and (c) signaling downstream of PKC in response to mitogenic stimulation. (2) Determine the effect of microgravity on PKC regulated genes which control monocyte differentiation, the initiation of apoptosis and cell cycle arrest (3) Determine if it is possible to restore agonist		

Task Description:	induced activation of individual PKC isoforms inhibited by microgravity by application of PKC isoform specific translocation enhancers and in turn determine whether these agents can restore expression of genes perturbed by microgravity. The experimental approach consists of stimulating PKC isoform activation in U937 cells using differentiation inducing agonists in microgravity and a 1.g on-board centrifuge. Gene chips will be used to identify the changes in the pattern of gene expression between microgravity, to identify which PKC isoform signaling pathways are affected by microgravity. Reverse transcriptase – polymerase chain reaction (Rt-PCR) will be used to confirm expression of key genes identified from the gene array data. The sub-cellular localization of PKC isoforms and expression of key surface activation markers will be determined in formaldehyde fixed samples. Collectively, these data will allow us to explore in detail the consequences of perturbed PKC signaling in microgravity at the level of gene expression and cell fate. We plan to analyze the expression of patterns of gene expression induced by differentiation inducing agonists in the U937 human monocyte cell line in microgravity and 1.g, to determine the functional effects of alterations in PKC signaling that occur in microgravity. The objectives of this research project are: (1) Fully characterize the effect of microgravity on, (a) the activation of key PKC isoforms (b) spatial distribution / redistribution of PKC isoforms within the cell and (c) signaling downstream of PKC in response to mitogenic stimulation. (2) Determine the effect of microgravity on PKC regulated genes which control monocyte differentiation, the initiation of apoptosis and cell cycle arrest (3) Determine if it is possible to restore agonist induced activation of individual PKC isoforms inhibited by microgravity by application of PKC isoform specific translocation enhancers and in turn determine whether these agents can restore expression of genes perturbed by microgravity. These studies will help us understand the changes in immunology that have been documented over the past 40 years of spaceflight. Using state of the are methods we will be able to identify specific microgravity sensitive signal pathways and gene inductions. Moreover, in understanding these parameters, we will be able to predict and possibly modify these microgravity disfunctions. Using gene chip and pathway studies from spaceflight we will be able to verify or deny the effectiveness of ground microgravity models (RPM and Rotating Vessel). Collectively, these data will allow us to explore in detail the mechanisms and consequences of perturbed PKC signaling in microgravity at the level of gene expression and cell fate.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This study investigates the mechanisms by which microgravity affects protein kinase C (PKC) signaling in monocytes and the functional consequences of these changes on downstream signaling events, gene expression and cell fate. This is important since many earthbound biological processes depend on this pathway and indeed the immune system itself is dependent on normal PKC signalling. We are analyzing the expression of patterns of gene expression induced by differentiation inducing agonists in the U937 human monocyte cell line in microgravity and 1.g, and are determining the functional effects of alterations in PKC signaling that occur in microgravity as well as on earth. Since all of Earths life forms have developed in a 1-g environment we will also learn if 1-g is necessary for full monocyte function.
Task Progress:	This experiment is ready for flight. We have made significant progress this year in definition of the flight experiment and developing protocols for the flight. Further funding is needed to complete baseline ground studies and to pay for gene chip analysis prior to the Kubik flight estimated for Oct, 2006. First we have developed our reatime rtpcr techniques in this laboratory and are able to verify chip data with qrtpcr. We have found that the PKC pathway gene induction with phorbol esters of TNF, ICAM, IL1b and cox-2 are significantlly inhibited induction of these genes under simulated microgravity- once the gene chip analysis is underway, we expect to see at least 20-30 genes to be dependent on a normal earth gravity vector. We have also developed flow cytometry methodology for this experiment to assess cell activation and cell cycle progression in a population of samples. We have tested flow cytometry protocols to quantify expression of early activation markers, including CD11c (an adhesion molecule, important in monocyte function during differentiation) and CD13 (a monocyte differentiation marker). Additionally we have tested propidium iodine staining as a method for quantifying the number of cells in each phase of the cell cycle using flow cytometry. U937 cells were stimulated to differentiate using 50ng/ml PDBu. We found that phorbol ester stimulation results in a significant upregulation of CD11c and CD13 in U937 monocytes. We have found changes in lymphocyte gene expression due to lack of gravity vector. Data from Affymetrix gene chips has been used to analyze the patterns and clustering of gene expression in the flight experiment. Using Affy Mass5 and GeneSpring we have analyzed gene expression from T-cells that were mitogen activated in a normal 1-g vector or in a random g-vector (random positioning machine, RPM). we have compared the gene expression of cells activated in a 1-g vector or in the RPM with no g-vector . The gene expression in are different, both in abundance and gene type. Many genes remain the same, only a few of the 35,000 human genes are affected by gravity. Most striking is that in the gravity vector, over 168 genes were induced over 4 fold within 4 hours of activation, while only 17 of those genes are induced the RPM activated samples. We have also started clustering the data to examine both the pattern of expression as well as beginning to examine the specific pathways that are induced by gravity. n=4 for each condition. As soon as the new budget arrives we will start the gene expression analysis for PKINASE differentiation using gene chip technology. Stability of RNA samples at ambient temperature (to simulate Soyuz Taxi flight and ISS KUBIK experiment) Most importantly, we have verified the viability of the RNA with sample stored at cabin temperature. Under flight conditions at ambient storage RNA from the U937 cells were processed and then triplicate samples were treated and then stored at –20oC for 10 days, or stored on benchtop at ambient temperature for 10 days before running RNA gel to check for purity. RNA samples stored at ambient had no visible degradation and are chip ready. We have refined this based on our preliminary data & current experiment design. The current experiment flow consists of 12 experiment cassettes, with 6 cassettes on the 1.g centrifuge & 6 cassette in microgravity, as described in figure 4 below. This experiment flow best accommodates the objectives of the experiment within the capabilities and constraints of the KUBIK facility and was developed based on data from our preliminary experiments.
Bibliography Type:	Description: (Last Updated: 03/10/2018)

Articles in Peer-reviewed Journals	Hughes-Fulford M "Signal transduction and mechanical stress" Science's STKE. Aug 31; (249):RE12 2004. PMID: 15353762 , Sep-2004
Articles in Peer-reviewed Journals	Hughes-Fulford M, Sugano E, Schopper T, Li C-F, Boonyaratanakornkit J, Cogoli A. "Early immune response and regulation of IL-2 receptor subunits" Cellular Signalling In Press 2005. , May-2005
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Presentation	M. Hughes-Fulford ""Microgravity and gene expression"" 2004 EB2004, Washington DC Apr-2004
Presentation	M Hughes-Fulford "Gene expression and microgravity" 2004 International IADR meeting Feb-2004
Presentation	M. Hughes-Fulford "How good is science aboard the shuttle and the space station" MIT Knight Journalism Fellows Oct-2004
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Presentation	M. Hughes-Fulford "Gravity regulates T-cell activation" Experimental Biology 2005 symposium Apr-2005